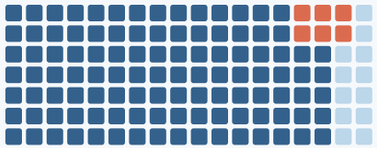
A Datasets

Three cohorts, two modalities. Each structured forget set is a rare biologically coherent subpopulation.

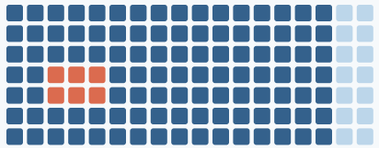
PBMC-33k
single-cell RNA-seq



33,088 cells · 2,000 HVGs

forget: 30 megakaryocytes

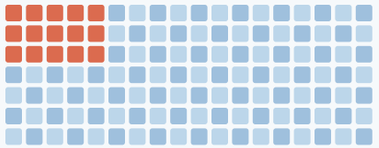
Tabula Muris
single-cell RNA-seq



41,647 cells · 2,000 HVGs

forget: 82 cardiac muscle

TCGA-BRCA
bulk RNA-seq



1,089 patients · 978 L1000

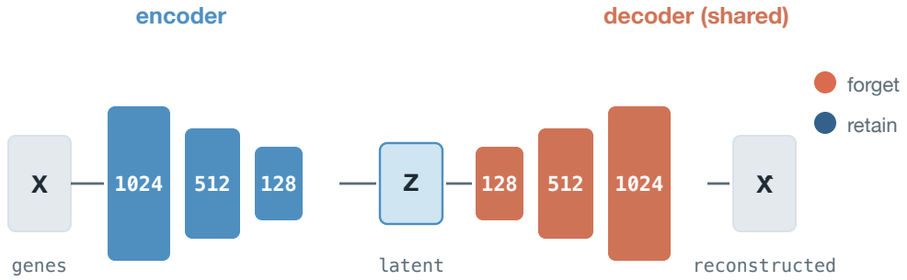
forget: 158 Basal subtype

retain cells · unseen (matched negatives) · forget set (structured)

train

B VAE

scVI-style encoder-decoder with shared output parameters.
Latent dim 32.

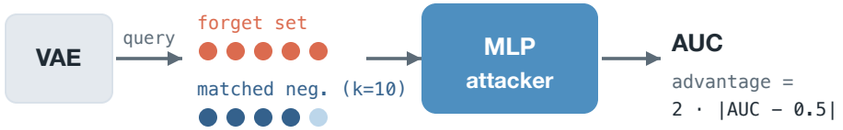


$L = -E[\log p(x|z)] + KL[q(z|x) \| N(0,I)]$

both pass through the same parameters

C Membership inference

Trained MLP attacker on 70-dim VAE latent features classifies forget vs. matched-negative samples.



Reference (PBMC, structured):

baseline AUC **0.79** retrain AUC **0.58** chance **0.50**

D Unlearning & evaluation

Nine post-hoc methods plus two training-time approaches and a DP-SGD (from scratch) baseline. Only retrain reaches the target region.

11 METHODS, GROUPED

Fine-tune	retain FT · grad ascent · Moon
Fisher-based	Fisher scrub · SSD
Distillation	SCRUB · contrastive latent
Adversarial	frozen critics · extra-gradient
Training-time	synthetic aug. · RAU
Scratch (baseline)	DP-SGD · full retrain

